

Problem-Solution Fit

Date	30 June2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Banks and Financial Institutions • Bank Credit Analysts • Financial Compliance Officers • Credit Card Applicants	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Review • Easy to understand • Slow • Human errors Rule-Based System • Simple implementation • Cannot learn from historical data Machine Learning System • Fast • Accurate • Scalable • Automated	9. PROBLEM ROOT / CAUSE [RC] • Manual verification process. • Increasing number of applications. • Complex financial evaluation. • Human decision-making limitations.
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2. PROBLEMS / PAINS

+ ITS FREQUENCY [PR]

- Manual credit card application review is time-consuming.
- Human errors may lead to incorrect approval decisions.
- Large volumes of applications increase processing time.
- High frequency of applications requires automation.

6.

CUSTOMER LIMITATIONS

EG. BUDGET, DEVICES

[CL]

- Limited processing time.
- Manual verification workload.
- Risk of inconsistent decisions.
- Limited availability of experienced analysts.

10. YOUR SOLUTION

[SL]

Develop a **Machine Learning-based Credit Card Approval Prediction System** using **Logistic Regression, Decision Tree, Random Forest, and XGBoost**. The system preprocesses applicant data, predicts approval or rejection in real time, integrates with a Flask web application, and is deployable on **IBM Cloud**, enabling faster, more accurate, and scalable credit approval decisions.

3. TRIGGERS TO ACT

[TR]

- Increase in daily credit card applications.
- Need for faster decision-making.
- Requirement for consistent approval criteria.
- Demand for automated banking solutions.

7. BEHAVIOR

+ ITS INTENSITY [BE]

- Reviews every application manually.
- Verifies applicant income and credit history.
- Makes approval/rejection decisions.

Intensity: High (performed continuously throughout the day).

4. EMOTIONS

BEFORE / AFTER [EM]

Before

- Stressed
- Time pressure
- Uncertain
- Overloaded

After

- Confident
- Satisfied
- Efficient
- Reliable

8. CHANNELS OF BEHAVIOR

[CH]

ONLINE

- Flask Web Application
- IBM Cloud
- Banking Portal

OFFLINE

- Bank Branch
- Customer Service Desk
- Printed Application Forms

